QFlow P6 Final Same-Board FPGA Results

H0–H4 Physical Comparison on Digilent Nexys3

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Generated from frozen, hashed repository evidence

Final result. P6 completed: 10/10 steps. Five independent H0–H4 bitstreams met the 100 MHz constraint and were cold-programmed into volatile FPGA SRAM. Across 420 physical UART records, all 420 rows and all 3,780 result fields exactly matched the frozen fixed-point model, with **zero mismatches**.

Board / FPGA	Digilent Nexys3 Rev B / Spartan-6 XC6SLX16-2-CSG324
Clock	100 MHz; one kernel cycle = 10 ns
Physical modes	H0, H1, H2, H3 and H4
Main proposed mode	H4 reinforcement-learned adaptive profile controller
H4 physical proof	84/84 rows, 756/756 fields, display 4084, LD0 ON, LD1 OFF
Latency boundary	Kernel cycles only; UART, host capture, JTAG and display refresh excluded

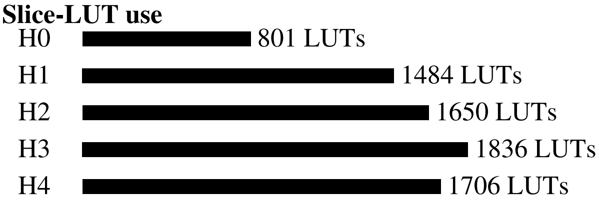
1. Same-board implementation and physical correctness

Mode	Controller	Fmax (MHz)	Cycles (normal)	Latency (μs)	Registers	LUTs	Slices	Exact fields
H0	Shortest distance	118.948	2.00	0.0200	493	801	301	756/756
H1	Fixed key-aware	107.538	201.76	2.0176	1000	1484	589	756/756
H2	Fixed QFlow	102.838	202.76	2.0276	1085	1650	649	756/756
H3	Rule adaptive	102.648	202.76	2.0276	1097	1836	645	756/756
H4	**RL adaptive**	**102.020**	**204.76**	**2.0476**	**1156**	**1706**	**677**	**756/756**

Table 1: Independent implementations using the same board, clock constraint, replay order, fixed-point contract and physical result shell.

Achieved Fmax (all exceed 100 MHz)

Slice-LUT use



H4 engineering trade-off versus H3. H4 uses **7.08% fewer LUTs** than H3, while registers change by 5.38%, occupied slices by 4.96%, and normal kernel latency by only 0.99% (two cycles, or 20 ns). Its Fmax changes by -0.61% but remains above the required 100 MHz.

2. Adaptive profile behavior

Mode	Controller	Profiles 0/1/2/3	Transitions	Dwell runs	Dwell range	Runtime Q update
H3	Threshold/rule adaptive	24/36/18/3	44	45	1-7	NO
H4	Reinforcement-learned adaptive	42/11/18/10	24	25	1-7	NO

Table 2: The three intentionally invalid rows are excluded; each mode has 81 valid/no-path profile decisions.

Main novelty demonstrated physically. H4 executed the frozen reinforcement-learned state-to-profile policy using all four profiles (42/11/18/10), produced 24 profile transitions and 25 dwell runs, and exactly matched the frozen model on every result field. Relative to H3, H4 made 45.45% fewer profile transitions, demonstrating a more stable dwell-aware learned profile sequence on this replay.

3. Held-out comparison and statistical interpretation

Comparison	Joint success	Mean diff.	Bootstrap 95% CI	Sign-test p	Paired d_z	Path changes
H4 vs H2	58	86.47	[-23.52, 228.22]	0.219	0.176	6
H4 vs H3	58	-78.05	[-242.53, 79.93]	0.453	-0.127	7

Table 3: Paired bottleneck-fidelity comparison on the 64-row held-out set. Positive differences favor H4.

Correct statistical claim. The H4-H2 mean fidelity difference is positive, while the H4-H3 difference is negative. In both comparisons the bootstrap 95% confidence interval crosses zero and the exact sign-test p-value exceeds 0.05. Therefore this replay supports physical correctness and adaptive execution, but not a statistically conclusive universal fidelity-superiority claim.

The held-out success rate is 90.625% for every mode when the two intentionally invalid rows are included. Raw scalar score is not compared across modes because adaptive modes can use different profile weights; comparing those scores directly would be invalid.

4. Thesis-ready contribution statement

1. **Fair hardware comparison:** H0–H4 were implemented and tested independently on the same Nexys3 FPGA rather than comparing FPGA latency against software-only literature results.
2. **Bit-exact physical validation:** 420/420 physical rows and 3,780/3,780 fields matched the frozen fixed-point golden model.
3. **Adaptive hardware behavior:** H3 proved threshold/rule switching; H4 proved reinforcement-learned adaptive profile selection with frozen dwell.
4. **Deterministic measurement:** all modes used the same 100 MHz clock, vector order, record format and cycle boundary.
5. **Efficient realization:** H4 met timing with 1,156 registers, 1,706 LUTs, 677 occupied slices, and no BRAM or DSP.

5. Claim boundary and limitations

Topic	Evidence-based statement	Status
Physical correctness	420 rows and 3,780 fields exactly match the frozen model	Supported
100 MHz operation	All five implementations have zero timing error and Fmax above 100 MHz	Supported
Learned adaptation	H4 follows a frozen learned state-to-profile and dwell policy	Supported
Online learning	No runtime Q-table update occurs on the FPGA	Do not claim
Universal fidelity superiority	Held-out confidence intervals include zero	Not established
Power / energy superiority	No common activity-calibrated H0–H4 evidence	Not measured
Photo/video evidence	OMITTED_NOT_ARCHIVED_IN_REPOSITORY	Explicitly labelled

Defense-safe wording. The evidence supports a reinforcement-learned adaptive FPGA controller that is physically correct, deterministic, timing-clean and resource feasible. It does not support online learning, universal network superiority, or power superiority.

6. Traceability

`same_board_comparison.csv`	Timing, cycles, resources and correctness
`heldout_pairwise_h4_statistics.csv`	Confidence intervals, sign tests and effect sizes
`profile_switching_dwell.csv`	Profile transitions and dwell
`physical_evidence_matrix.csv`	Programming, UART hashes, displays and LEDs
`power_energy_status.csv`	Explicit power/energy omission